

Matteo Tagliavini

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After earning my Master's Degree in Computer Engineering, with a specialization in Data Engineering and Analytics, I built a hybrid path across data, AI and robotics. During my studies I worked at a Robot System Integration company, an experience that kept me close to the world of robotics and that also inspired my research thesis: the integration of a collaborative robot with an omnidirectional mobile base and the development of an admittance control to make human-robot interaction more natural. I am looking for a stimulating and highly technological work environment, oriented towards growth, where I can make a concrete contribution to the development solutions and applications with real-world impact

Professional Experience

- October 2022 - Present **R&D Developer at Ideativa Srl, Modena**, Full-time.
Specialized in the development and integration of industrial and collaborative robotics solutions with brands such as ABB and FANUC, I work on R&D activities focused on Robotics Engineering and offline programming for production process optimization. I implement 2D/3D vision systems for advanced handling and assembly applications. I also play an active role in the Educate & Train area through the creation of the internal Academy and the definition of the company's technical-educational roadmaps.
- October 2022 - Present **Robotics Instructor & Technical Expert at Academy ITS Biomedicale, Mirandola**.
- March 2023 - Present **Robotics Instructor & Technical Expert at IAL Emilia Romagna, Modena**.
- April - July 2021 **Intern Student at Ideativa Srl, Modena**, Curricular internship.
Development of a pick&place solution with the ABB YuMi cobot integrating 3D vision and AI for component recognition, dataset creation and training, camera-robot calibration and Python software integration for automated execution.

Education

- 2021 - 2025 **University of Modena and Reggio Emilia, "Enzo Ferrari" Department of Engineering**, Master's Degree in Data Engineering and Analytics.
 - Title: Physical Human-Mobile Robot Interaction for Collaborative Transportation
 - Final Grade: 110/110
 - Graduation Date: 16/07/2025
- 2016 - 2021 **University of Modena and Reggio Emilia, "Enzo Ferrari" Department of Engineering**, Bachelor's Degree in Computer Engineering.
 - Title: Implementation of a robotic component picking solution using a vision system and integration of artificial intelligence algorithms
 - Final Grade: 90/110
 - Graduation Date: 21/10/2021
- 2011 - 2016 **Scientific High School (Applied Sciences), Alessandro Volta, Sassuolo**.
 - Diploma Grade: 74/100
 - Diploma Type: Italian diploma

Publications

- August 2023 **Online Motion Planning for Safe Human–Robot Cooperation Using B-Splines and Hidden Markov Models**, 2023-08-18 | *MDPI Journal article*
Authors: Giovanni Braglia; Matteo Tagliavini; Fabio Pini; Luigi Biagiotti.
Article proposing an online motion planning algorithm for safe human-robot cooperation, capable of handling unpredictable human movements by adapting the robot's trajectory and speed in real time through a combination of B-splines and Hidden Markov Models, with experimental validation on a Franka Emika Panda cobot.
- September 2026 **A Micro-Macro Manipulation Framework for Human-Mobile Robot Co-Manipulation Tasks**, September 2026 | *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*
Authors: Davide Tebaldi; Matteo Tagliavini; Fabio Pini; Luigi Biagiotti.
Proposes a control strategy for human–mobile manipulator co-manipulation based on the micro/macro paradigm, separating low-frequency movements at the mobile base from high-frequency ones at the robotic arm to enhance performance and interaction quality. Experimentally validated, demonstrating clear benefits in energy consumption, human effort, and execution time

Licenses and Certifications

- May 2025 **Winner of ABB RoboCup 2025.**
- February 2025 **RO13DL Roboguide FANUC.**
- February 2025 **RO09 iR Vision FANUC.**
- February 2025 **RO06 DCS FANUC.**
- February 2025 **RO04 Advanced Programming FANUC.**
- February 2025 **RO03 Basic Programming FANUC.**
- May 2024 **Winner of ABB RoboCup 2024.**

Skills

- Robotics RAPID (ABB), TP (FANUC), RoboStudio, RoboGuide
- Development Python, C/C++, Java, SQL, Git, Bash, L^AT_EX
- Tools VS Code, Eclipse, SQL Server Management Studio, Overleaf
- Languages Italian, English
- Soft skills Teamwork, Problem solving, Autonomy, Adaptability, Communication

Autorizzo il trattamento dei miei dati personali presenti nel CV ai sensi dell'art. 13 d. lgs. 30 giugno 2003 n. 196 - "Codice in materia di protezione dei dati personali" e dell'art. 13 GDPR 679/16 - "Regolamento europeo sulla protezione dei dati personali".

